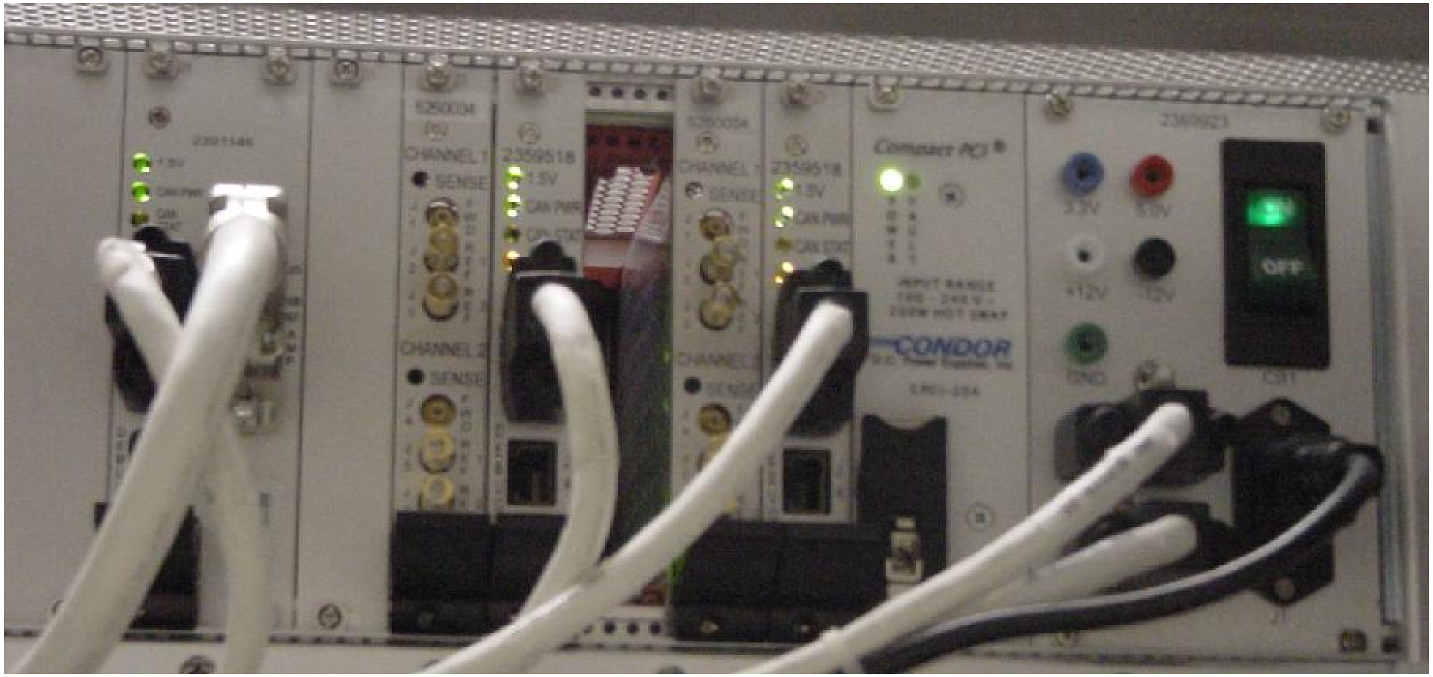
Amplifier Support Controller (ASC)



The ASC, Amplifier Support Controller, resides in the upper rear of the RFS Cabinet. It is in Forward Production Excite III systems, it will not be part of an upgrade to Excite III

- The ASC replaces the SSM power monitoring and SSM RF amplifier interface functionality.
- It provides amplifier interface (Control signals) to Host for both Narrow and Broadband RF amplifiers
- Contains UPM – Universal Power Monitor hardware.
- It communicates to the system through CAN copper cable link.
- The ASC is HART compatible and all the circuit boards are reported in the HARdware Tracker log.
- The power supply board has test points that allow for easy check using a DVM.
- It is possible to emulate out the boards associated with the ASC chassis in the mgd_stage directory.
- The ASC does require calibration when any of the hardware in the chain is replaced. However it is not necessary to calibrate during a new installation. The calibration is done in the factory and the calibration files are supplied on a CDROM that is shipped with the RFS cabinet.

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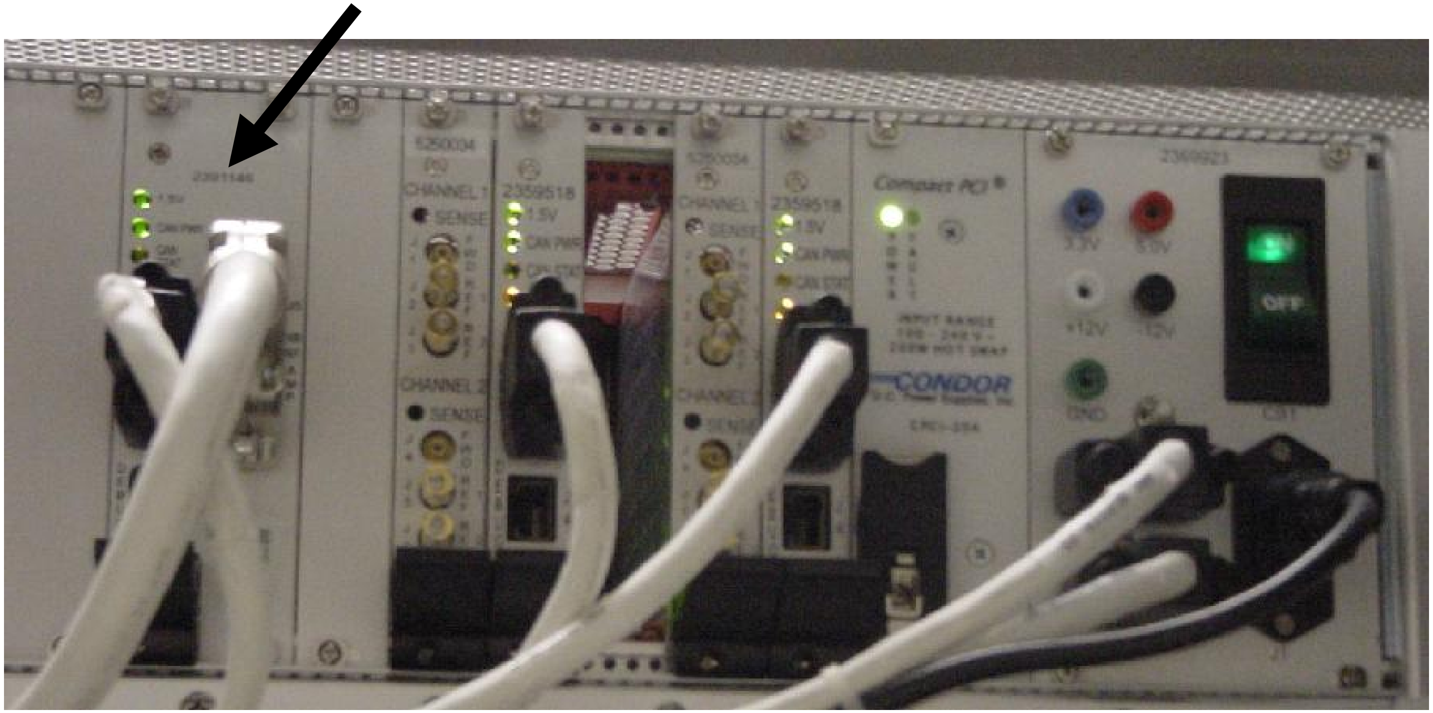
RF Interface board



This board permits the host to control and monitor the operation of the RF SRFD2 or MKS amplifier. The board controls both the Narrowband RF amp or the Broadband (MNS) amp.

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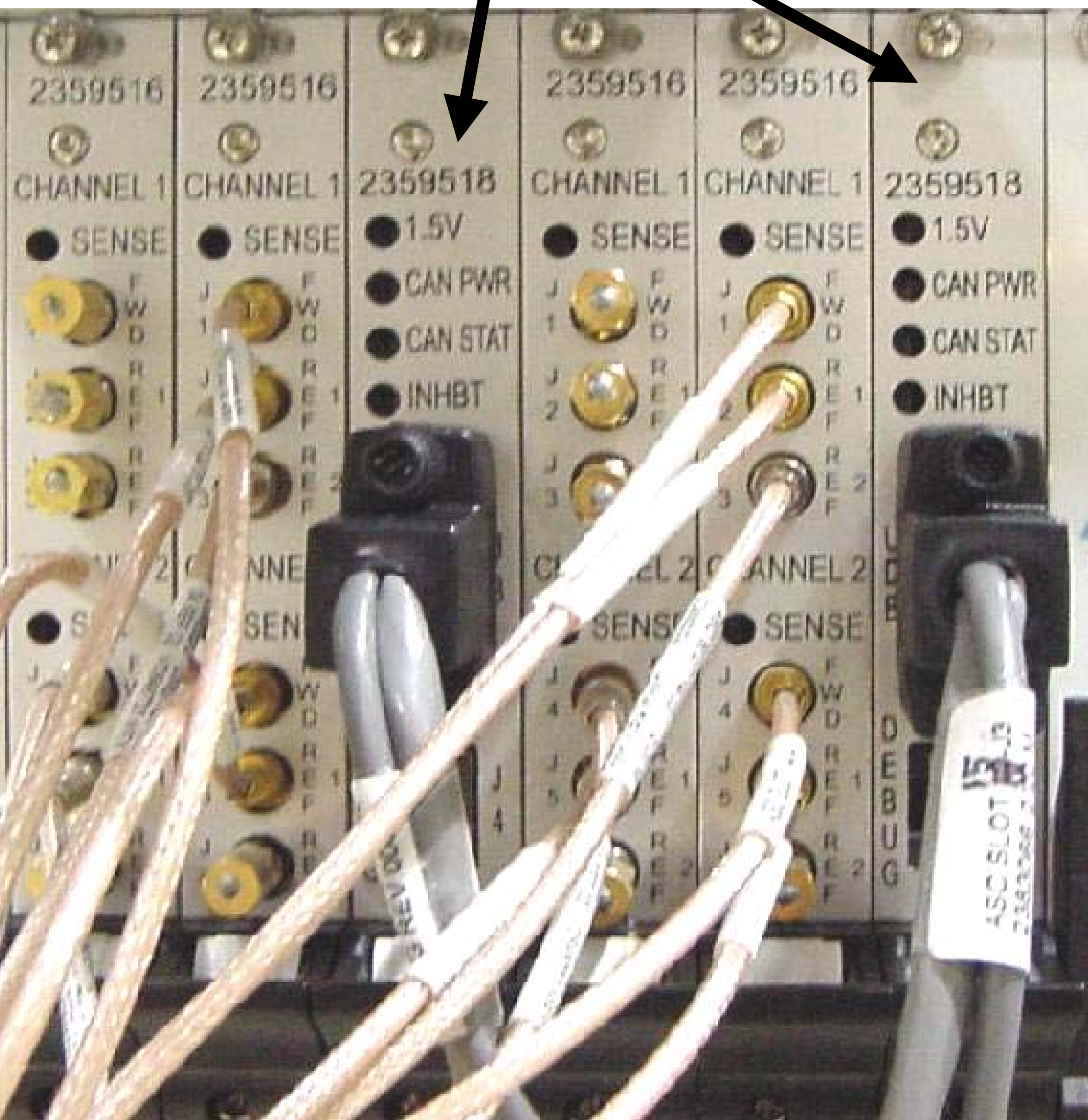
Processor boards



The Processor boards are part of the Universal Power Monitor. The Processor Boards monitor power monitor data from the Detector Boards and then report this to the system. The ASC provides redundant UPM monitoring so that if the hardware in section 1 fails then the hardware in section 2 can still detect a fault. This explains why there are 2 Processor Boards in each ASC Chassis. Both Processor Boards are identical and can be swapped during troubleshooting. The Detector Boards are installed directly to the left of each Processor Board. Each Processor Board can control up to 3 Detector Boards but the normal configuration is only 1 Detector Board per Processor Board. This picture in the slide also shows the 2nd Detector Board that is installed when the customer has the optional MultiNuclear Spectroscopy (MNS) hardware. Diagnostics are available for the Processor Boards and they also are reported in the HART Log.

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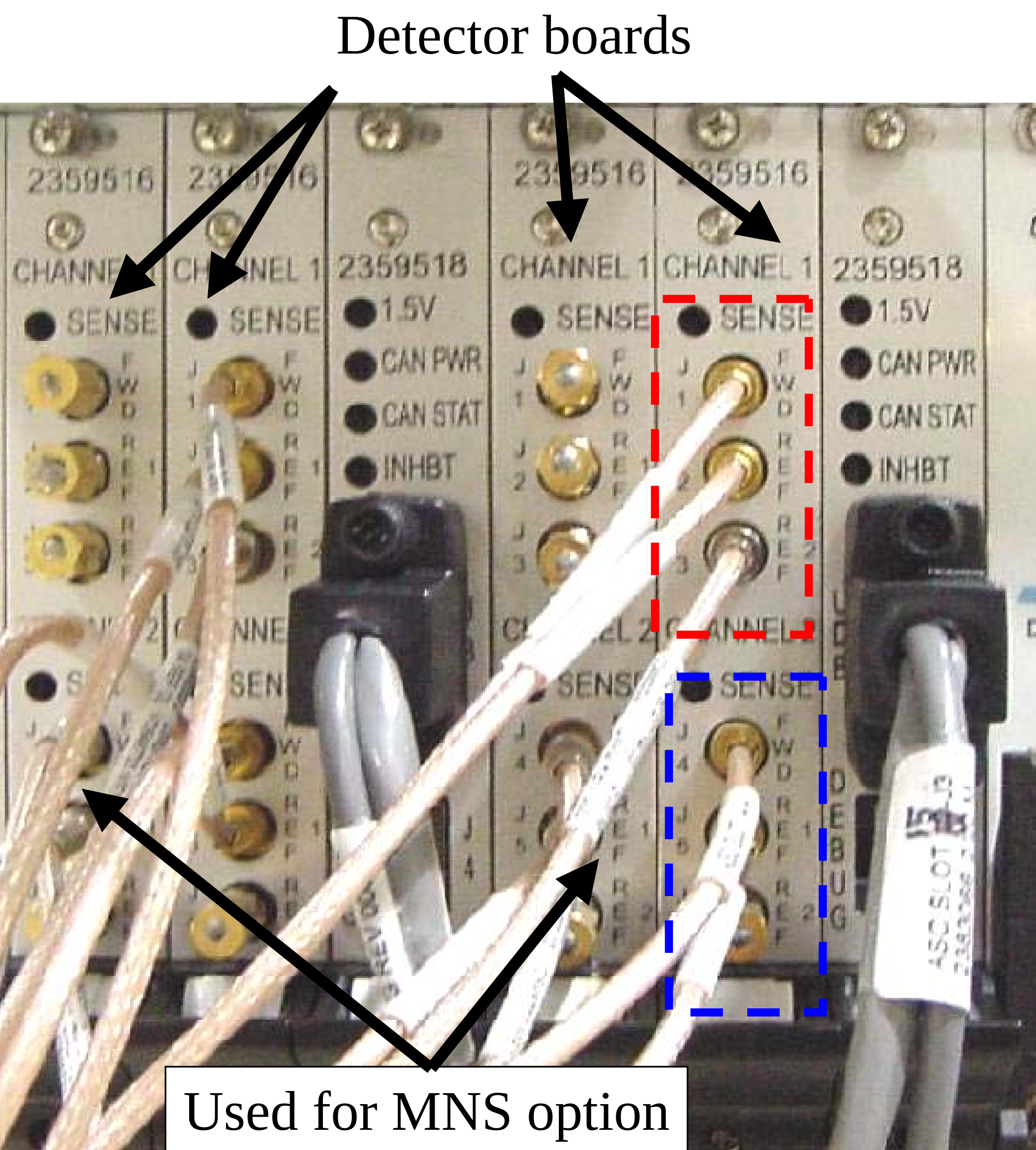

Detector Board 1 monitors head and body forward power in 1.5T and head and body forward and reflected power in 3.0T. The upper **red** box contains Channel 1, the Body monitoring section, while the lower **blue** box contains Channel 2, the Head monitoring section. The sampled head and body RF signals are provided from the white cables which route the signals from the connector ports located on the rear of the SRFD2 RF amplifier. Notice the white cable connected to the port labeled “REF 2” in the upper boxes. 3.0T systems will have a cable connected to this port. This was intended to provide sample RF signal from the 3.0T Quadrature Hybrid Splitter but it is not used. A 50 ohm terminator must terminate all other unused ports. This is seen from the picture. Diagnostics are available for the Detector Boards and they also are reported in the HART Log